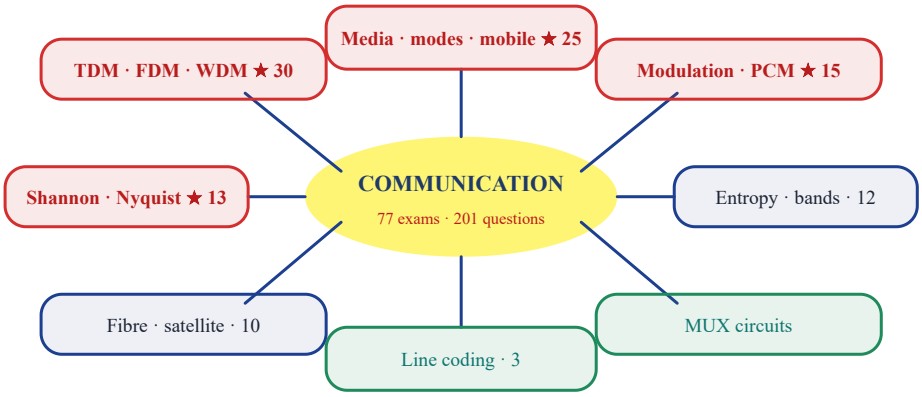
Communication Theory

every communication question asked in BCS · NTRCA · BPSC · bank · BTRC · BUET-set papers, solved in 29 sheets



↑ the number is how many different exams asked that topic (red = most asked)

How to read a question card

> **Q:** The question, as printed in the paper (typos fixed).
>
> ✎ **Asked in:** the exams it was taken from, e.g. 45th BCS CSE (2024) · BPSC AP 2021

No dated exam → the card names its source (suggestion sheet, bank IT viva).

Then: answer → green = worked result → trap.

Most asked

① TDM vs FDM (7 exams)
② $C = B \log_2(1 + S/N)$
③ Nyquist rate of $\sin(2100\pi t)$
④ PCM: blocks, pros/cons
⑤ FDM + guard bands
⑥ TDM frame numericals
⑦ AM / FM bandwidth

Syllabus coverage

"Spectral analysis: Fourier series, **sampling function**, power spectrum …; **Information theory: entropy, information rate, Shannon's theorem, channel capacity**; Analog communication system: **different modulations** …; Digital modulation: different standard modulation schemes; Pulse and digital signals: PAM, **PCM, DM, ADM**; **Multiplexing: TDM, FDM**, multiple-access network: TDMA, FDMA, **CDMA**".

> **Legend:** **yellow** = key term · **red circle** = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = worked answer

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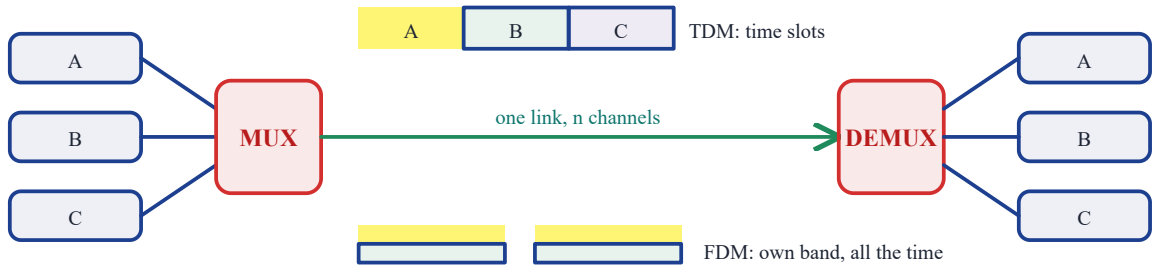
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1 TDM vs FDM: the most-asked question

Q1: What is multiplexing in data communication? Write the differences between TDM and FDM.

✎ **Asked in:** 45th BCS CSE (2024) · BPSC AP 2021 · 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · 36th BCS (2016)

Definition: **Multiplexing** sends several signals over **one shared link** at the same time: a MUX combines n inputs onto one line, a DEMUX separates them at the far end. It is used because one link's capacity is usually far larger than one user needs.



Point	TDM	FDM	WDM
Shares	**time**: each input gets a slot	**frequency**: each input gets a band	wavelength (colour of light)
Signal type	digital	analog	optical (analog in concept)
Medium	wire, fibre, radio	coax, radio (TV, FM)	optical fibre only
Separation	sync bits / guard time	guard bands	guard wavelength gap
Needs	**tight synchronisation**	filters, carriers	prisms / gratings
Problem	empty slots waste capacity	cross-talk, intermodulation	costly lasers
Example	T1/E1 lines, GSM slots	radio, cable TV, AMPS	DWDM backbone, SEA-ME-WE

Memory trick: **T**DM = **T**urns in **T**ime; **F**DM = everyone talks at once on a different **F**loor (band); **W**DM = FDM done with **W**avelengths of light.

Q2: Differentiate among TDM, FDM and WDM. How does synchronous TDM work? Write about synchronous and statistical TDM.

✎ **Asked in:** Combined Bank SO(IT) 2023 · RAKUB ANSE 2023

Synchronous TDM: every input owns a **fixed slot** in every frame, whether it has data or not. Frame = one slot per input (+ framing bits). Output rate = n × input rate. **Empty slot = wasted capacity.**

Statistical TDM: slots are given **only to inputs with data**, so each slot carries an **address**. No fixed frame; link rate can be below n × input rate. Used by packet networks.

Exam trap: "FDM is for analog, TDM for digital" is the standard line, but FDM can carry modulated digital data too (Wi-Fi OFDM). The 49th BCS MCQ asks TDM's **main disadvantage over FDM**: **complex synchronisation**.

Q3: What is multiplexing? Describe the different types of multiplexing, with examples, and explain time-division multiplexing in detail.

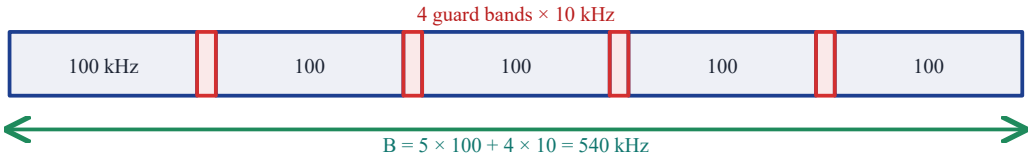
✎ **Asked in:** Combined Bank AME/AHE 2023 · SPCBL AME 2021 · SPCBL SAME 2022

1. **FDM**: each signal modulates its own carrier; bands are separated by guard bands (radio, TV, 1G AMPS).
2. **WDM**: several light wavelengths on one fibre, combined by a prism / grating (DWDM: 40–160 λ).
3. **TDM**: synchronous (fixed slot per input) and statistical (slot on demand, with an address). Interleaving unit = bit, byte or block; a framing bit marks each frame.
4. **CDM / CDMA**: every user sends in the whole band at the same time, separated by orthogonal codes (3G).

2 FDM numericals: guard bands

Q4: Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

✎ **Asked in:** Combined Bank SO(IT) 2026



→ $B_{min} = N \cdot B + (N - 1) \cdot G = 500 + 40 =$ **540 kHz**

Memory trick: Fence posts: N channels need only **N − 1** guard bands (none at the two edges).

Q5: Solve an FDM numerical: frequency allocation and required bandwidth (the figures are not legible on the surviving paper).

✎ **Asked in:** BSCL AE(CSE) 2026

The usual version: three 4-kHz voice channels share a link of 12 kHz, 20–32 kHz. Each channel modulates its own carrier and keeps the upper band: **ch 1 → 20–24 kHz, ch 2 → 24–28 kHz, ch 3 → 28–32 kHz**; the receiver separates them with band-pass filters and demodulates. No guard band here: if one is required, add (N − 1)·G as in Q4.

Q6: AMPS uses two bands: 824–849 MHz for sending and 869–894 MHz for receiving. Each user has a bandwidth of 30 kHz in each direction. How many people can use their cell phones simultaneously?

✎ **Asked in:** JBL ANA 2022

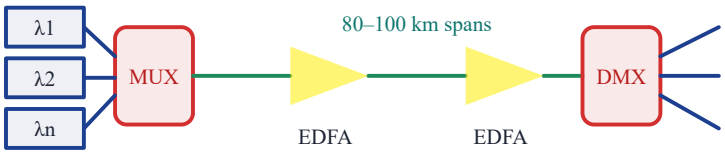
→ each band = 25 MHz; 25 MHz ÷ 30 kHz = 833.3 → AMPS uses **832 channels**; 42 are control channels → **790 users** at once (one channel in each band per user)

Exam trap: Do not add the two bands (50 MHz ÷ 30 kHz): a call needs one 30-kHz channel **in each direction**, so the bands give the same count. If the paper ignores control channels, write 833 and say why.

Q7: Where are the low-loss transmission windows of silica optical fibre, and which window is the most popular? Draw a long-haul WDM transmission system.

✎ **Asked in:** BTCL AM(Technical) 2021

Window	Loss
850 nm	≈ 2–3 dB/km
1310 nm	≈ 0.35 dB/km
1550 nm	**≈ 0.2 dB/km**



1550 nm is the most popular: lowest loss, and **erbium-doped fibre amplifiers (EDFA)** amplify the whole 1530–1565 nm C-band at once, so dozens of DWDM channels cross thousands of km with only optical repeaters.

3 TDM frame numericals

Three rules solve every synchronous-TDM question: ① **frame duration = duration of one input unit** (each input sends one unit per frame) · ② output slot = frame ÷ n · ③ output rate = n × input rate (+ framing bits × frame rate).

Q8: Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot and (d) the duration of a frame.

✎ **Asked in:** BB AME 2019 · also on an AHE/ANE suggestion set

→ (a) 1/1000 s = **1 ms** · (b) 4 × 1 kbps = **4 kbps** · (c) 1/4000 s = **250 µs** · (d) 4 slots × 250 µs = **1 ms**

Q9: The figure shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate and (d) the output frame rate.

✎ **Asked in:** BTCL AM(Technical) 2021 · JBL SO(ANE) 2020

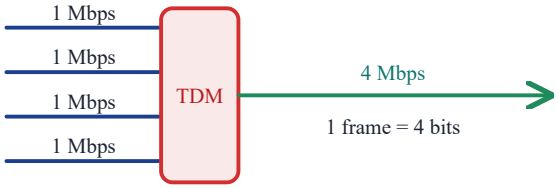


Figure (textbook version): four inputs at 1 Mbps.
(a) input bit = 1/1 Mbps = **1 µs**
(b) output bit = 1/4 Mbps = **0.25 µs**
(c) output rate = 4 × 1 Mbps = **4 Mbps**
(d) frame rate = input rate = **1,000,000 frames/s**

Q10: Two channels, one at 190 kbps and the other at 180 kbps, are multiplexed using pulse-stuffing TDM with no synchronisation bits. Find (a) the frame size in bits, (b) the frame rate, (c) the frame duration and (d) the data rate.
✎ **Asked in:** RAKUB ME 2021

→ pulse stuffing raises the 180-kbps channel to 190 kbps · (a) **2 bits** (1 per channel) · (b) **190,000 frames/s** · (c) 1/190,000 = **5.26 µs** · (d) 2 × 190 kbps = **380 kbps**

Q11: A TDMA system has 8 transmitter–receiver pairs. Each is sampled at 8 kHz, giving 16 bits per sample. Two synchronisation bits are added to each frame. Calculate the data rate of the TDMA link.
✎ **Asked in:** BTCL AM(Technical) 2021

→ frame = 8 × 16 + 2 = 130 bits; frames/s = 8000 → R = 130 × 8000 = **1.04 Mbps**

Q12: In a TDM system eight signals of 2 kHz each are sampled 25% above the Nyquist rate; the quantisation error must not exceed 1% of the signal amplitude. Find the bit rate (recalled wording).
✎ **Asked in:** NESCO AM(ICT) 2021

→ fs = 1.25 × 2 × 2 kHz = 5 kHz · error ≤ Δ/2 = 1% of peak m_p, $\Delta = 2m_p/L \rightarrow L \geq 100 \rightarrow$ 128 levels, **n = 7** · R = 8 × 5000 × 7 = **280 kbps**

Q13: How long does it take to send a file of 500,000 bits from host A to host B over a circuit-switched network where all links are 1.536 Mbps, each link uses TDM with 32 slots, and it takes 800 ms to establish the end-to-end circuit?
✎ **Asked in:** Combined Bank AHE 2021 · also on the Petrobangla ANE suggestion sheet

→ one circuit = 1.536 Mbps ÷ 32 = 48 kbps · transfer = 500,000 ÷ 48,000 = 10.42 s · total = 10.42 + 0.8 = **11.22 s**

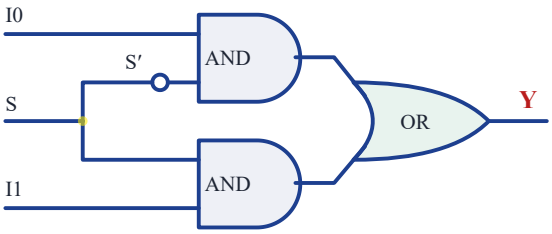
Exam trap: A circuit gets **one slot**, i.e. 1/32 of the link, never the full 1.536 Mbps; and the set-up time is added once. Suggestion-sheet variant (16 slots, 1.5 Mbps, no set-up): 500,000 ÷ 93,750 = 5.33 s.

Drill: A 2-Mbps TDM channel carries 500-bit frames. (a) Sampling rate per user = 2,000,000 ÷ 500 = **4000 samples/s**. (b) 8-bit samples + 4 sync bits per frame → (500 − 4) ÷ 8 = **62 users**. ✎ **Source:** Petrobangla ANE suggestion sheet

4 Multiplexer circuits

These are digital-logic questions; the papers file them next to TDM. A **multiplexer (MUX)** is a data selector: 2^n inputs, n select lines, one output. A **demultiplexer (DEMUX)** sends one input to one of 2^n outputs.

Q14: Draw the logic circuit of a two-input multiplexer. Compare multiplexer and demultiplexer.
✎ **Asked in:** 45th BCS CSE (2024)



Y = S′·I0 + S·I1

Point	MUX	DEMUX
Lines	many → **one**	**one** → many
Selects	which input passes	which output gets it
Size	$2^n : 1$	$1 : 2^n$
Built from	AND-OR	decoder + data
Use	parallel → serial, TDM sender	serial → parallel, TDM receiver

Applications ✎ **Asked in:** Combined Bank Officer(IT) 2026: MUX picks one of several sensors for one ADC, or one of several registers onto a bus; DEMUX routes a bus to one of several devices (memory chip select, keyboard scanning).

> **Q15:** Design a 4-to-1 multiplexer and explain its working.
> ✎ **Asked in:** 46th BCS ICT · DESCO SAE 2022 · 16th NTRCA CSE (2019)

Y = S1′S0′I0 + S1′S0 I1 + S1 S0′I2 + S1 S0 I3: four 3-input AND gates (each gets one data input and one select combination) feed a 4-input OR; the select code S1S0 = 00, 01, 10, 11 passes I0, I1, I2, I3.

> **Q16:** Draw and write the output equation for an 8 × 1 multiplexer with an enable input E (E = L → output 0; E = H, S2S1S0 = LLL → D0 … HHH → D7).
> ✎ **Asked in:** Combined Bank SO 2020

→ Y = E·(S2′S1′S0′D0 + S2′S1′S0D1 + S2′S1S0′D2 + S2′S1S0D3 + S2S1′S0′D4 + S2S1′S0D5 + S2S1S0′D6 + S2S1S0D7)

> **Q17:** Draw a logic circuit for a 2-to-4 demultiplexer.
> ✎ **Asked in:** Rupali Bank ANE 2021

→ Y0 = D·S1′S0′ · Y1 = D·S1′S0 · Y2 = D·S1S0′ · Y3 = D·S1S0 (two inverters, four 3-input AND gates)

> **Q18:** Design a three-input XOR, f = w1 ⊕ w2 ⊕ w3, using 2-to-1 multiplexers.
> ✎ **Asked in:** Dhaka WASA AME(Network) 2025

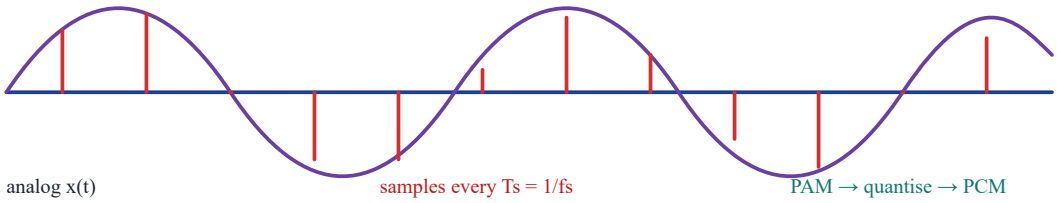
→ MUX 1: select w1, I0 = w2, I1 = w2′ → g = w1 ⊕ w2 · MUX 2: select w3, I0 = g, I1 = g′ → f = g ⊕ w3. Without inverters, a third MUX with I0 = 1, I1 = 0 makes g′.

MCQ	Answer
a. Parallel-to-serial converter: (a) decoder (b) encoder (c) demultiplexer (d) multiplexer ✎ **Asked in:** BUET IICT April 2021	**(d) multiplexer**
b. 2:1 MUX, output fed back to I0, data on I1, select S: (a) D flip-flop (b) D latch (c) half-adder (d) demultiplexer ✎ **Asked in:** BCIC AP 2025	**(b) D latch**
c. 8086 address lines multiplexed with data: (a) 8 (b) 32 (c) 64 (d) 16 ✎ **Asked in:** 49th BCS ICT (2025)	**(d) 16**

Exam trap: (b) is a **latch**, not a flip-flop: while S = 1 the output follows D (level-sensitive); S = 0 holds it. (c) The 8086 multiplexes **AD0–AD15**: 16. "8" is the 8088.

5 The sampling theorem

Nyquist sampling theorem: A signal band-limited to f_{max} is fully recovered from its samples if $\mathbf{f_s \geq 2f_{max}}$. Nyquist rate = $2f_{max}$; Nyquist interval = $1/(2f_{max})$. Sampling slower causes **aliasing**: high frequencies fold back as false low ones.



Q19: Determine the Nyquist sampling rate and the Nyquist sampling interval for x(t) = sin(2100πt).
✎ **Asked in:** 38th BCS (2018)

1. Compare with sin(2πft): 2πf = 2100π → $\mathbf{f_{max} = 1050\ Hz}$
2. Nyquist rate = 2 × 1050 = **2100 samples/s**
3. Nyquist interval = 1/2100 s ≈ **0.476 ms (476 µs)**

Exam trap: The coefficient of t is **ω = 2πf**, not f. Students write f = 2100 Hz and get 4200 samples/s. Some printed answers write "f = d/2" without a result: always finish with the number and the unit.

Q20: A message of bandwidth 20 kHz and amplitude 10 V is sampled at a rate 150% higher than the Nyquist rate and quantised uniformly; the bit rate is 1.6 Mbps. Find (i) bits per sample, (ii) the SQNR, (iii) the step size.
✎ **Asked in:** 46th BCS ICT

1. Nyquist rate = 40 kHz; "150% higher" = 40 × (1 + 1.5) = **100 kHz**
2. (i) n = 1.6 Mbps ÷ 100 kHz = **16 bits/sample**
3. (ii) sine input: SQNR = 1.76 + 6.02n = 1.76 + 96.32 = **98.1 dB**
4. (iii) Δ = 2A / 2^n = 20 V / 65536 ≈ **305 µV**

Exam trap: "150% **higher** than" = 2.5 × Nyquist; "150% **of**" = 1.5 × Nyquist (60 kHz → 26.7 bits/sample, not a whole number, which shows it is the wrong reading).

MCQ	Answer
a. g(t), band-limited to B Hz, has Nyquist samples 0 everywhere except 1 at t = 0. Then: (a) impulse at t = 0 (b) odd periodic (c) g = 0 for $\|t\| > 1/2B$ (d) g = 0 at t = n/2B, $n \neq 0$ ✎ **Asked in:** BUET M.Sc. April 2021	**(d) g = 0 at t = n/2B, n ≠ 0**
b. $\sin(2\pi Bt)$ is band-limited to B Hz; $\sin^n(2\pi Bt)$ to: (a) 2^nB (b) B/n (c) nB (d) 2nB ✎ **Asked in:** BUET M.Sc. April 2021	**(c) nB**

(a) g(t) is the **sinc** pulse, $\sin(2\pi Bt)/(2\pi Bt)$: zero at every other sampling instant, but not zero for all $|t| > 1/2B$.

(b) Multiplying in time = convolving spectra, so each power adds B: $\sin^2$ has 2B, $\sin^n$ has nB.

Trick: "Twice the **highest** frequency", never twice the bandwidth of a band-pass signal by default.

Drill: 300–3400 Hz voice, PCM, error ≤ ±1% of full scale, range −1 V to +1 V: $\Delta/2 = 0.02$ V → $\Delta = 0.04$ V → $L = 2/0.04 = 50$ → **6 bits**; at the Nyquist rate 6.8 kHz → 40.8 kbps. ✎ **Source:** tail of an unidentified paper in a coaching compilation

6 Channel capacity: two formulas

Nyquist (noiseless): $\mathbf{R = 2B \log_2 L}$

B = bandwidth, L = signal levels. More levels → more bits, until noise stops you.

Shannon (noisy): $\mathbf{C = B \log_2(1 + SNR)}$

SNR as a **ratio**: $SNR = 10^{dB/10}$. The ceiling no coding can beat.

Q21: A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity of this channel?
✎ **Asked in:** Combined Bank AP 2023 · RPGCL AM(ICT) 2022

→ $C = 3000 \times \log_2(1 + 3162) = 3000 \times 11.63 =$ **34,881 bps ≈ 34.88 kbps** (3162 = 35 dB)

Exam trap: One printed answer writes "SNR = 3162 dB": 3162 is the **power ratio** (= 35 dB). Putting 3162 dB into the formula is meaningless.

Q22: Find the channel capacity of a teleprinter channel with 300 Hz bandwidth and an SNR of 3 dB.
✎ **Asked in:** MRA AME 2022

→ $SNR = 10^{0.3} = 1.995$ → $C = 300 \times \log_2(2.995) = 300 \times 1.583 =$ **474.8 bps** (one printed copy misprints 30 dB in the "given" line)

Q23: For a noisy channel, the signal-to-noise power ratio is 0.1 and the bandwidth is 2.4 GHz. What is the channel capacity in Mbps?
✎ **Asked in:** 46th BCS ICT

→ C = 2.4 × 10^9 × $\log_2(1.1)$ = 2.4 × 10^9 × 0.1375 = **330 Mbps** (SNR below 1 still gives capacity: spread-spectrum works this way)

MCQ	Answer
a. Gaussian channel, SNR = 15, B = 16 kHz: C (kbps) = (a) 32 (b) 64 (c) 18 (d) 128 ✎ **Asked in:** 49th BCS ICT (2025)	**(b) 64**
b. B = 1 MHz, SNR = 63: bit rate = (a) 4 Mbps (b) 6 Mbps (c) 8 Mbps (d) 10 Mbps ✎ **Asked in:** BPSC AME 2019	**(b) 6 Mbps**
c. B = 200 kHz, 4 signal levels, max data rate: (a) 400 kbps (b) 800 kbps (c) 1000 kbps (d) 1200 kbps ✎ **Asked in:** Combined Bank SO(IT) 2021	**(b) 800 kbps**

→ (a) 16k × $\log_2$16 = 64 kbps · (b) 1M × $\log_2$64 = 6 Mbps · (c) Nyquist: 2 × 200k × $\log_2$4 = 800 kbps

Q24: Using the Nyquist theorem, a channel can be sampled 12 million times per second; four-level signals give 2 bits per sample. What is the total data rate? (6-MHz noiseless channel.)
✎ **Asked in:** NESCO AM(ICT) 2021

→ R = 2 × 6 MHz × $\log_2$4 = **24 Mbps**

Q25: Signal power 10 mW, noise power 1 μW: find SNR and SNR in dB.
✎ **Asked in:** MGMCL AM(ICT) 2022

→ SNR = 10 × 10^{-3} / 10^{-6} = **10,000** → 10 $\log_{10}(10^4)$ = **40 dB**. Noiseless channel: SNR → ∞.

Drill: 265 kbps over a noiseless 20-kHz channel: 265,000 = 2 × 20,000 × $\log_2$L → $\log_2$L = 6.625 → L ≈ 98.7 → use **128 levels** (7 bits) in practice. ✎ **Source:** worked example in a coaching compilation

Memory trick: Shannon = **S**NR inside the log (**S**hannon **S**ays noise); Nyquist = **N**umber of levels inside. Use Shannon for the limit, Nyquist for the levels.

7 Bit rate, baud and decibels

Key relations: bit rate N = r × baud S, where r = $\log_2$L bits per signal element · dB = 10 $\log_{10}(P_2/P_1)$ · gains in dB **add** · −3 dB = half power · 10 dB = ×10.

Q26: An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements are needed?
✎ **Asked in:** Sonali Bank Officer(IT) 2021

→ r = 8000 / 1000 = **8 bits per signal element** → L = 2^8 = **256 different signal elements**

Q27: A digitised TV picture uses 480 × 500 pixels, each with one of 32 intensity values; 30 pictures are sent per second. Find the source rate R.
✎ **Asked in:** BB AME 2023

→ 480 × 500 = 240,000 pixels × $\log_2 32$ = 5 bits × 30 /s = **36 Mbps**

Q28: How many 8-bit characters can be sent per second over a 9600-baud serial link in asynchronous mode with one start bit, eight data bits, two stop bits and one parity bit?
✎ **Asked in:** Combined Bank SO(IT) 2022

→ bits per character = 1 + 8 + 1 + 2 = 12 → 9600 ÷ 12 = **800 characters/s** (binary line: 1 baud = 1 bit/s)

Q29: (a) A signal is attenuated by 10 dB; what is its final power if it was originally 5 W? (b) A signal passes through three cascaded amplifiers of 4 dB gain each: total gain? (c) On a 5-kbps channel, how long does a 100,000-bit frame take to send?
✎ **Asked in:** BTCL AE(CSE) 2017

→ (a) −10 dB = ÷10 → **0.5 W** · (b) 4 + 4 + 4 = **12 dB** (power × 15.85) · (c) 100,000 ÷ 5000 = **20 s**

Q30: What is 3 dB? How can you turn a converter into an inverter?
✎ **Asked in:** BTRC AD(Technical) 2019

3 dB = a power ratio of 2 (10 log 2 = 3.01): −3 dB is **half power** (0.707 of voltage). The **3-dB bandwidth** is the band between the two half-power points of a filter or amplifier. **Converter → inverter:** a fully controlled (thyristor) rectifier works as a **line-commutated inverter** when its firing angle is above 90° and a DC source of the right polarity drives it: power then flows from DC to the AC side.

Q31: A link uses 16-QAM; the signal operates at 2400 baud. Calculate the data rate in bps.
✎ **Asked in:** RAKUB ANSE 2026

→ 16-QAM: $\log_2 16$ = 4 bits per baud → 4 × 2400 = **9600 bps**

MCQ	Answer
a. Same bit rate: QPSK symbol time T_Q vs BPSK T_B: (a) $0.25T_B$ (b) $4T_B$ (c) $0.5T_B$ (d) $2T_B$ ✎ **Asked in:** 49th BCS ICT (2025)	**(d) $2T_B$**

QPSK carries 2 bits per symbol, so at the same bit rate each symbol lasts **twice** as long and needs half the bandwidth.

Drill: 8-bit samples at 8 kHz: bit rate = 8 × 8000 = **64 kbps**; on a binary (2-level) line the baud rate is also 64 kbaud. This is one PCM voice channel (DS0). ✎ **Source:** AHE/ANE suggestion set

Exam trap: Baud counts **signal changes**, bps counts **bits**; they are equal only for 2-level signalling. dB values add only when they are gains or losses; dBm is an absolute power (0 dBm = 1 mW).

8 Modulation and AM

Q32: What is modulation? Why is it necessary? What are modulation and demodulation? Write five uses.

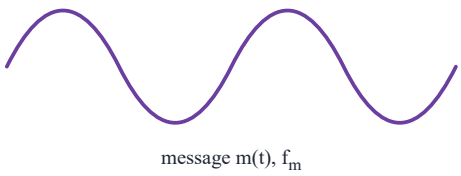
✎ **Asked in:** BPSC AE 2022 · BCSCL NE 2017

Definition: **Modulation** varies a property (amplitude, frequency or phase) of a high-frequency **carrier** in step with the message. **Demodulation** recovers the message at the receiver.

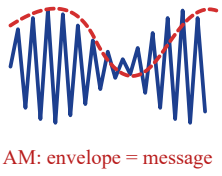
1. **Practical antenna size**: length ≈ λ/4; 3 kHz audio needs a 25-km antenna, 100 MHz needs 75 cm
2. **Multiplexing**: different carriers share one medium (FDM)
3. **Range**: high frequencies radiate and travel further
4. **Noise**: FM / digital schemes resist noise
5. **Matching**: shift the signal into the band the channel passes

Uses: ① AM/FM radio broadcast ② TV ③ mobile phones (GMSK, QAM) ④ modems / DSL ⑤ Wi-Fi, satellite and radar links.

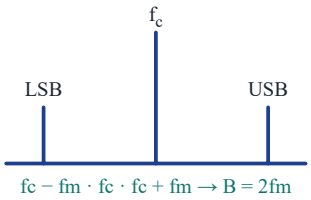
Memory trick: Why modulate? "**A**ntenna, **M**ultiplex, **R**ange, **N**oise" = **AMRN**.



message m(t), f_m



AM: envelope = message



$f_c - f_m \cdot f_c \cdot f_c + f_m \rightarrow B = 2f_m$

MCQ	Answer
a. Carrier 1 MHz, message 10 kHz, AM bandwidth: (a) 5 kHz (b) 10 kHz (c) 20 kHz (d) 2 MHz ✎ **Asked in:** 49th BCS ICT (2025)	**(c) 20 kHz**
b. Carrier 300 kHz, message 3 kHz, AM bandwidth: (a) 300 Hz (b) 3 kHz (c) 6 kHz (d) 600 Hz ✎ **Asked in:** DoICT ANE 2019	**(c) 6 kHz**
c. In amplitude modulation: (a) signal amplitude changes (b) signal frequency changes (c) carrier amplitude changes (d) carrier frequency changes ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(c) carrier amplitude changes**

Exam trap: AM bandwidth = **2 × f_m**, independent of the carrier: the 1-MHz carrier is a distractor. SSB halves it to f_m.

Q33: A DSB full-carrier AM wave is v = 20(1 + 0.7 sin 4500t) sin 45000t volt. Find (a) the carrier frequency, (b) the modulating frequency, (c) the modulation index, (d) the lower and (e) the upper side-band frequency.

✎ **Asked in:** BTRC DAD 2021

→ ω_c = 45000 rad/s → (a) $f_c = 45000/2\pi$ = **7162 Hz** · (b) $f_m = 4500/2\pi$ = **716.2 Hz** · (c) m = **0.7** · (d) LSB = 7162 − 716 = **6446 Hz** · (e) USB = **7878 Hz**

Q34: Compare AM and FM in terms of bandwidth, noise performance and power efficiency.
✎ **Asked in:** 47th BCS ICT

Point	AM	FM
Varies	carrier amplitude	carrier frequency
Bandwidth	**$2f_m$** (10 kHz broadcast)	**$2(\Delta f + f_m)$** (200 kHz broadcast)
Noise	poor: noise adds to amplitude	**good**: limiter removes amplitude noise
Power	≥ 2/3 of power in the carrier	constant envelope, all power useful
Range / cost	long (sky wave), simple receiver	line of sight, complex receiver

9 FM numericals

FM toolkit: $\Delta f = k_f \times A_m$ · modulation index **$\beta = \Delta f / f_m$** · **Carson's rule** $B = 2(\Delta f + f_m) = 2f_m(\beta + 1)$ · $\beta < 1$ narrowband ($B \approx 2f_m$).

Q35: A sinusoidal modulating wave of amplitude 5 V and frequency 2 kHz is applied to an FM generator with a frequency sensitivity of 40 Hz/V. Calculate the frequency deviation, the modulation index and the bandwidth.
✎ **Asked in:** BOF AP 2022

→ $\Delta f = 40 \times 5$ = **200 Hz** · $\beta = 200 / 2000$ = **0.1** (narrowband FM) · $B = 2(200 + 2000)$ = **4.4 kHz** ($\approx 2f_m$ = 4 kHz)

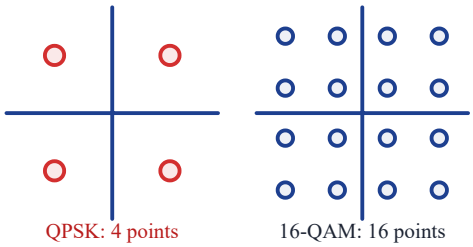
Q36: A frequency-modulated voltage wave is $x(t) = 10 \cos(6.28 \times 10^8 t + 5 \sin 1256t)$. (The paper is cut off after the equation; the usual parts are below.)
✎ **Asked in:** 47th BCS ICT

→ $f_c = 6.28 \times 10^8 / 2\pi \approx$ **100 MHz** · $f_m = 1256 / 6.28$ = **200 Hz** · β = **5** · $\Delta f = 5 \times 200$ = **1 kHz** · $B = 2(1000 + 200)$ = **2.4 kHz** · power in 1 Ω = $10^2/2$ = **50 W**

Exam trap: In the equation the number in front of sin is **β**, not Δf; and 6.28 ≈ 2π, so divide it away. FM power never changes with β.

10 Digital modulation

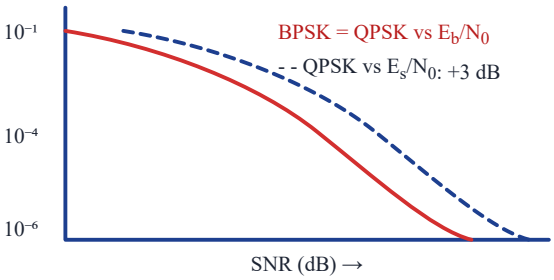
Scheme	Varies	bits/symbol
ASK	amplitude	1
FSK / MSK / **GMSK**	frequency	1
BPSK	phase (0°, 180°)	1
QPSK	phase (4 states)	2
16-QAM	amplitude + phase	**4**
64-QAM	amplitude + phase	6



MCQ	Answer
a. Most bandwidth-efficient: (a) BPSK (b) QPSK (c) FSK (d) 16-QAM ✎ **Asked in:** 49th BCS ICT (2025)	**(d) 16-QAM**
b. Modulation used in GSM: (a) MSK (b) FSK (c) ASK (d) GMSK ✎ **Asked in:** 49th BCS ICT (2025)	**(d) GMSK**

Q37: Draw the bit error rate (BER) vs signal-to-noise ratio curves of QPSK and BPSK.

✎ **Asked in:** NWPGCL AM(ICT) 2024



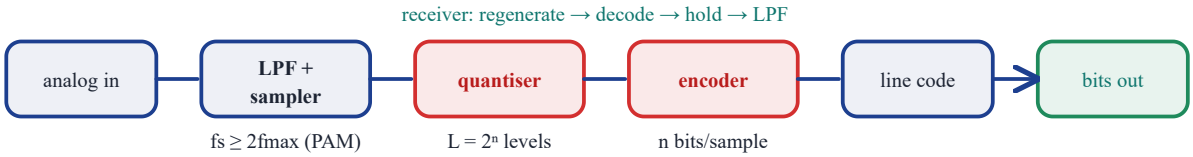
$P_b = Q(\sqrt{(2E_b/N_0)})$ for **both**. Values: 0 dB → 7.9×10^{-2}, 4 dB → 1.3×10^{-2}, 8 dB → 1.9×10^{-4}, 10 dB → 3.9×10^{-6}.

→ Against E_b/N_0 the curves **coincide**; against symbol SNR (E_s/N_0) QPSK sits **3 dB to the right**, because each symbol carries 2 bits. QPSK gives double the data rate in the same bandwidth for the same BER.

11 Pulse code modulation

Q38: What is modulation? Describe the pulse code modulation (PCM) technique. State the advantages and disadvantages of PCM.

✎ **Asked in:** 38th BCS (2018) · also on the 46th & 47th BCS suggestion sheet



Steps. ① band-limit and **sample** at $f_s \geq 2f_{max}$ ② **quantise** each sample to one of $L = 2^n$ levels ③ **encode** each level as n bits ④ send as a line code. Bit rate = $n \cdot f_s$ (voice: 8×8000 = **64 kbps**).

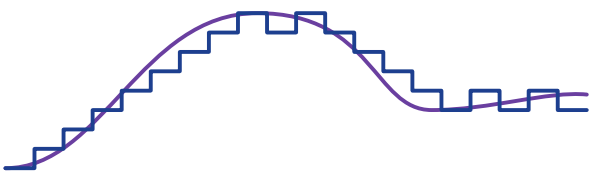
Advantages	Disadvantages
noise immune: repeaters regenerate	needs far more bandwidth (64 k vs 4 kHz)
easy TDM, encryption, storage	quantisation noise always present
error detection / correction	complex coder; needs sync

Q39: A PCM system has a uniform quantiser with step size 2 V; the input is a sinusoid of amplitude 10 V. Calculate the signal-to-quantisation-noise ratio in dB and the number of bits per sample.
✎ **Asked in:** BTCL AM(Technical) 2021

1. Range = $2 \times 10 = 20$ V → L = 20 / 2 = 10 levels → n = $\lceil \log_2 10 \rceil$ = **4 bits**
2. Signal power = $A^2/2 = 50$; noise = $\Delta^2/12 = 4/12 = 0.333$
3. SQNR = 50 / 0.333 = 150 → 10 log 150 = **21.8 dB**

Exam trap: 1.76 + 6.02n (= 25.8 dB for n = 4) assumes the levels fill the full range with $L = 2^n = 16$; here the step is fixed at 2 V, so only 10 levels are used: work from $A^2/2 \div \Delta^2/12$.

12 Delta modulation: one bit per sample



DM sends 1 if the signal is above the staircase, 0 if below; the staircase moves one step Δ. **ADM** changes Δ with the slope (bigger on steep parts, smaller on flat parts), reducing **both** noises. **DPCM** sends the quantised difference with several bits.

MCQ	Answer
a. Higher slope-overload and granular noise: (a) DPCM (b) DM (c) ADM (d) PCM ✎ **Asked in:** 49th BCS ICT (2025)	**(b) DM**
b. True for delta modulation: (a) larger step ↓ granular noise (b) smaller step ↑ slope overload (c) smaller step ↑ granular noise (d) larger step ↑ slope overload ✎ **Asked in:** BUET M.Sc. April 2021	**(b) smaller step ↑ slope overload**

Memory trick: **Small** step can't **climb** (slope overload); **big** step can't **sit still** (granular). ADM = adaptive step fixes both.

Hamming drills: 4 data bits need r parity bits with $2^r \geq m + r + 1$ → **r = 3**, code (7,4). Hamming **distance** = number of positions that differ: 10101 ⊕ 11110 = 01011 → **3**. ✎ **Source:** Petrobangla ANE suggestion sheet · worked example in a coaching compilation

13 Line coding

Q40: What is line coding? Define the line-coding techniques and mention their categories.

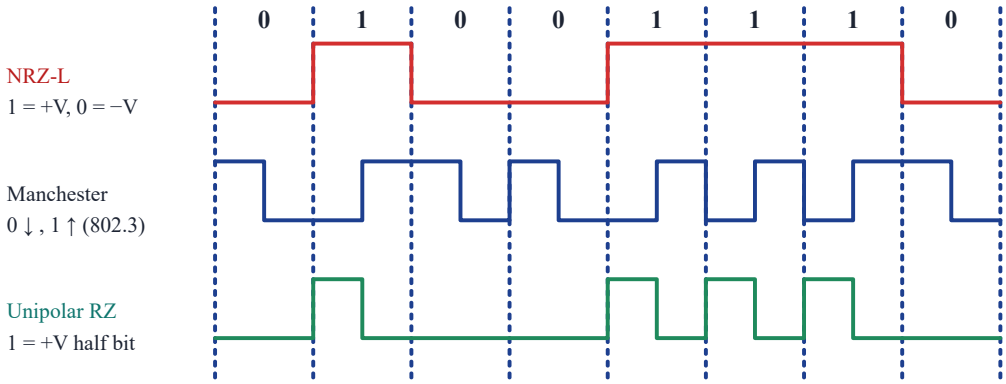
✎ **Asked in:** SPCBL AME 2021

Definition: **Line coding** converts a sequence of bits into a digital signal (voltage levels) for the wire. A good code has no DC component, carries its own clock (self-synchronising), uses little bandwidth and detects some errors.

Category	Schemes	Note
Unipolar	NRZ, RZ (one level + zero)	DC component, rarely used
Polar	NRZ-L, NRZ-I, RZ, **Manchester**, **differential Manchester**	Manchester: Ethernet 10 Mbps; diff. Manchester: Token Ring
Bipolar	AMI, pseudoternary (B8ZS, HDB3 scrambling)	T1 / E1 lines
Multilevel	2B1Q, 8B6T, 4D-PAM5	ISDN, 100Base-T4, Gigabit
Multitransition	MLT-3	100Base-TX

Q41: We want to transmit the binary string 01001110. Show the resulting signal using (i) NRZ-L, (ii) Manchester, (iii) unipolar RZ.

✎ **Asked in:** BPSC AE 2022



Exam trap: State your convention: some books draw NRZ-L with 0 = +V, and Manchester per G. E. Thomas is the inverse of IEEE 802.3. Any consistent convention earns full marks; an unstated one does not. Manchester always has a **mid-bit transition**.

Q42: A communication numerical on NRZ and RZ line coding (recalled wording).

✎ **Asked in:** Teletalk Executive(Technical) 2015

→ the usual ask is bandwidth for N bps: NRZ changes at most once per bit → $B_{min} \approx N/2$ (1 Mbps → 500 kHz); RZ and Manchester change twice per bit → $B_{min} \approx N$ (1 Mbps → 1 MHz), the price of self-clocking.

MCQ	Answer
a. 1110000100110101 in 4B/5B: max consecutive zeros in the output: (a) 1 (b) 4 (c) 2 (d) 3 ✎ **Asked in:** BUET M.Sc. April 2021	**(d) 3**
b. Read a differential Manchester waveform (figure not reproduced): (a) 10000111 (b) 01000100 (c) 01111000 (d) 10111011 ✎ **Asked in:** BUET M.Sc. April 2021	**see the rule below**

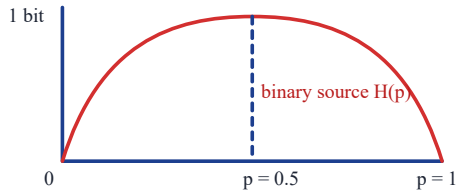
→ (a) 1110→11100, 0001→01001, 0011→10101, 0101→01011 → 11100 01001 10101 01011: longest run of 0s = **3** (4B/5B never allows more than 3). (b) Rule: a transition at the **start** of a bit = **0**, no transition = **1**; the mid-bit transition is only the clock.

14 Entropy

Q43: Explain the concept of entropy in information theory. How does it quantify the uncertainty of a random message source?

✎ **Asked in:** 47th BCS ICT

Definition: The information in a symbol of probability p is $I = \log_2(1/p)$ bits: rare symbols carry more. **Entropy** is the **average information per symbol**: $\mathbf{H = -\Sigma\, p_i \log_2 p_i}$ bits/symbol. It is largest ($\log_2 M$) when all M symbols are equally likely (most uncertain) and 0 when one symbol is certain. Information rate $R = r \times H$ (r = symbols per second).



certain event → H = 0

equally likely → $H = \log_2 M$ (max)

MCQ	Answer
a. The expression for entropy: (a) $\Sigma P_i \log_2 P_i$ (b) $-\Sigma \log_2 P_i$ (c) $-\Sigma P_i \log_2 P_i$ (d) $\Sigma P_i \log_2(P_i/(1+P_i))$ ✎ **Asked in:** 49th BCS ICT (2025)	**(c) $-\Sigma P_i \log_2 P_i$**

Q44: A discrete memoryless source emits A, B, C, D with probabilities 0.5, 0.25, 0.15, 0.10. (i) Calculate the entropy in bits per symbol. (ii) If one symbol is sent every millisecond, find the information rate.

✎ **Asked in:** 47th BCS ICT

Symbol	p	$\log_2(1/p)$	$p \cdot \log_2(1/p)$
A	0.50	1	0.5000
B	0.25	2	0.5000
C	0.15	2.737	0.4105
D	0.10	3.322	0.3322

→ (i) H = **1.743 bits/symbol** · (ii) r = 1000 symbols/s → R = 1000 × 1.743 = **1743 bps**

Q45: A source sends one of four symbols A, B, C, D as 2-bit pulses (00, 01, 10, 11); each bit lasts 5 ns. P(A) = 1/5, P(B) = 1/4, P(C) = 1/4, P(D) = 3/10. Calculate the average rate of information transmission.

✎ **Asked in:** DPDC AM(ICT) 2025

1. $H = 0.2 \log_2 5 + 2 \times 0.25 \log_2 4 + 0.3 \log_2(10/3) = 0.464 + 1.000 + 0.521 =$ **1.985 bits/symbol**
2. symbol time = 2 × 5 ns = 10 ns → r = **10^8 symbols/s**
3. $R = r \times H = 10^8 \times 1.985 =$ **1.985×10^8 bps ≈ 198.5 Mbps**

Exam trap: The raw bit rate is 200 Mbps (2 bits every 10 ns), but the **information** rate is lower (198.5 Mbps) because the symbols are not equally likely. Equal probabilities would give exactly 2 bits/symbol.

Memory trick: Halves give whole bits: p = ½ → 1 bit, ¼ → 2 bits, ⅛ → 3 bits. Use this to check a table quickly.

15 Frequencies, bands and channels

Band	Range	Use
MF	300 kHz–3 MHz	AM radio
HF	3–30 MHz	short wave
VHF	**30–300 MHz**	FM, TV
UHF	300 MHz–3 GHz	mobile, Wi-Fi
SHF	3–30 GHz	satellite, microwave
EHF	30–300 GHz	mmWave, 5G

MCQ	Answer
a. VHF range: (a) 3–30 MHz (b) 30–300 MHz (c) 300 MHz–3 GHz (d) 3–30 GHz ✎ **Asked in:** 49th BCS ICT (2025)	**(b) 30–300 MHz**

Trick: Every band is a ×10 step that starts with **3**: 3 k, 30 k, 300 k, 3 M, 30 M (VHF starts here), 300 M …

Q46: What is the difference between audio frequency and radio frequency?

✎ **Asked in:** BTRC DAD 2021 · BTRC DAD(Technical) 2019

Audio frequency (AF): 20 Hz–20 kHz, what the ear hears; travels as sound (pressure waves) and cannot radiate efficiently from an antenna. **Radio frequency (RF)**: about 3 kHz–300 GHz electromagnetic waves that radiate from an antenna and carry AF as modulation. AF = message, RF = carrier.

Q47: Dhaka radio broadcasts on medium wave at "630 Hz". If radio waves travel at 3×10^8 m/s, what is the wavelength?

✎ **Asked in:** BTRC AD(Technical) 2019 · BTRC SAD 2021

→ $\lambda = c/f = 3 \times 10^8 / 630 =$ **476,190 m** as printed; the station's real frequency is **630 kHz** → $\lambda =$ **476.2 m** (medium wave = hundreds of metres)

MCQ	Answer
b. Period 10 ms: frequency = (a) 10 Hz (b) 20 Hz (c) 50 Hz (d) 100 Hz ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(d) 100 Hz**
c. Bursts of 697 Hz and 1477 Hz: key pressed = (a) 1 (b) 2 (c) 3 (d) 4 ✎ **Asked in:** Combined Bank SO(IT) 2021	**(c) 3**

(c) **DTMF**: each key = one row tone + one column tone. Rows 697 / 770 / 852 / 941 Hz; columns 1209 / 1336 / **1477** Hz. 697 = top row (1 2 3), 1477 = third column → **3**.

> **Q48:** Differentiate low-pass and band-pass channels with examples.
> ✎ **Asked in:** 15th NTRCA ICT

Low-pass: passes 0 to f_1 (starts at zero) → baseband digital signals go straight on it (a LAN cable). **Band-pass**: passes f_1 to f_2, not starting at zero → the signal must be **modulated** to fit (a radio or telephone voice channel, 300–3400 Hz). A low-pass channel is usually one wide medium used by one pair of devices; a band-pass channel is a slice of a shared medium.

> **Q49:** What is the main benefit of a broadband transmission system compared with baseband?
> ✎ **Asked in:** Rupali Bank ANA 2023

Baseband puts one digital signal on the whole bandwidth of the cable (Ethernet). **Broadband** modulates signals onto separate carriers, so **many channels travel at once** (FDM: cable TV, DSL, cable Internet) and over longer distances with amplifiers.

> **Q50:** What is channel hardening, and what does it imply for frequency allocation (in OFDMA, for example)?
> ✎ **Asked in:** BTRC DAD(Technical) 2019

In **massive MIMO**, with many base-station antennas the fading on the individual paths averages out, so the effective channel gain of each user becomes almost **constant** (deterministic) over time and frequency. Then OFDMA need not chase the best sub-carriers per user (frequency-selective scheduling); every user can get the whole band, with fewer pilots and simpler power control.

> **Exam trap:** $\lambda = c/f$: kHz → hundreds of metres, MHz → metres, GHz → centimetres. A wavelength of 476 km for a broadcast station signals a unit slip.

16 Simplex, half-duplex, full-duplex

> **Q51:** Write the differences between simplex, half-duplex and duplex communication with examples.
> ✎ **Asked in:** 15th NTRCA ICT

Mode	Direction	Capacity	Example
Simplex	one way only	whole channel one way	keyboard → CPU, TV / radio broadcast
Half-duplex	both ways, **one at a time**	whole channel, taking turns	walkie-talkie, CB radio
Full-duplex	both ways **at once**	split / two paths	telephone, mobile call

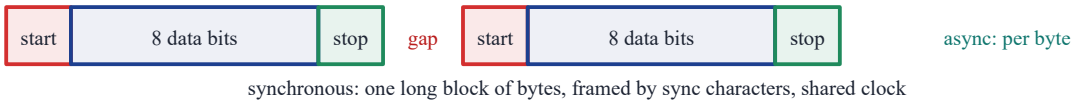
MCQ	Answer
a. Mode used during a phone conversation: (a) simplex (b) full-duplex (c) half-duplex (d) multiplex ✎ **Asked in:** 49th BCS CSE (2025)	**(b) full-duplex**
b. New phone: a user cannot talk while the other talks, but either side may start: (a) simplex (b) half-duplex (c) full-duplex (d) none ✎ **Asked in:** BUET M.Sc. April 2021	**(b) half-duplex**
c. Which mode uses serial communication? (a) duplex (b) half duplex (c) simplex (d) all ✎ **Asked in:** BCC AP 2021	**(d) all**

Exam trap: (b) "one-way at a time, from either side" is **half-duplex**, not simplex: simplex can never reverse. (c) Serial / parallel describes how bits travel; direction is a separate property, so all three modes can be serial.

17 Synchronous vs asynchronous transmission

Q52: Write the differences between synchronous and asynchronous transmission, with advantages and disadvantages. What is the problem of synchronous transmission, and how does asynchronous transmission solve it?

✎ **Asked in:** RAKUB ANSE 2023 · 15th NTRCA Computer Operation (2019) · WZPDCL AE(CSE) 2019



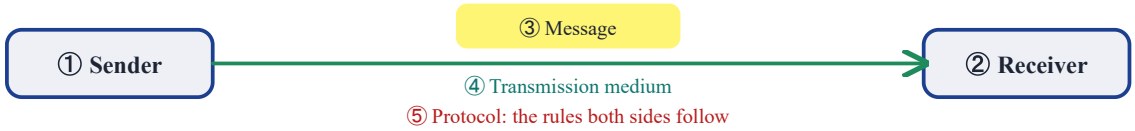
synchronous: one long block of bytes, framed by sync characters, shared clock

Point	Asynchronous	Synchronous
Unit	one character, start + stop bits	a block / frame of many bytes
Clock	re-synchronised by each start bit	shared or recovered clock
Gaps	random idle time between bytes	none: continuous stream
Overhead / speed	≥ 20% (2 of 10 bits); slow	small; fast
Cost / use	cheap; keyboard, RS-232, UART	needs buffers; LAN, SONET, HDLC

The problem with synchronous mode: sender and receiver must keep their clocks locked for the whole block; a small drift over thousands of bits misplaces every later bit, and the receiver must buffer the block. **The asynchronous fix:** each character brings its own start bit, so the receiver **re-locks every 10 bits**; drift never accumulates, at the cost of the start/stop overhead.

Q53: Name and define the five components of a data communication system, with a diagram.

✎ **Asked in:** BPSC AP 2023



Message = the data (text, voice, video); **sender / receiver** = the devices; **medium** = the physical path (cable, fibre, radio); **protocol** = agreed rules (format, timing, error handling). Without a protocol two connected devices exchange signals but no information.

Q54: What is data transmission speed? Discuss its types.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

The number of bits sent per second (bps). **Narrow band**: up to about 300 bps (telegraph) · **Voice band**: up to about 9600 bps–56 kbps (telephone line, modem) · **Broadband**: 1 Mbps and above (fibre, DSL, coax, microwave, satellite).

18 Transmission media

Q55: What types of transmission media are used for data communication? Explain their advantages and disadvantages. What is attenuation? Distinguish twisted pair, coaxial cable and optical fibre in a table.

✎ **Asked in:** Combined Bank SO(IT) 2024 · Rupali Bank ANA 2023 · 15th NTRCA ICT

Guided (wired): twisted pair, coaxial, optical fibre. **Unguided** (wireless): radio waves, microwave, infrared, satellite. **Attenuation** = loss of signal power with distance, in dB (or dB/km); amplifiers / repeaters restore it.

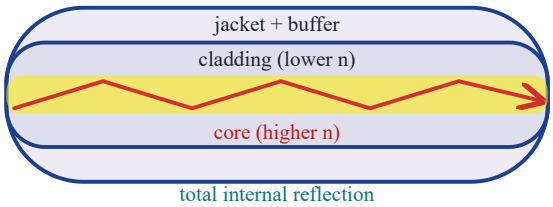
Point	Twisted pair	Coaxial	Optical fibre
Carries	electrical signal	electrical signal	**light**
Bandwidth	low (≤ 10 Gbps short)	medium	**highest**
Attenuation	high: repeater ~100 m	medium	lowest: 0.2 dB/km
EMI / tapping	affected (twist reduces)	shield helps	immune, hard to tap
Cost / install	cheapest, easy	moderate	costly, skilled splicing
Use	LAN (Cat-6), telephone	cable TV, CCTV	backbone, submarine

MCQ	Answer
a. A guided medium: (a) radio wave (b) microwave (c) coaxial cable (d) infrared ✎ **Asked in:** 49th BCS CSE (2025)	**(c) coaxial cable**
b. Cable for an analog CCTV camera: (a) UTP (b) coaxial (RG-59) (c) fibre (d) HDMI ✎ **Asked in:** Combined Bank Officer(IT) 2026	**(b) coaxial (RG-59)**
c. Needs line of sight: (a) microwave (b) fibre (c) twisted pair (d) coaxial ✎ **Asked in:** BPSC AP 2022	**(a) microwave**
d. A disadvantage of optical fibre: (a) resists corrosion (b) immune to tapping (c) unidirectional light (d) none of these ✎ **Asked in:** Combined Bank AHE/AE(IT) 2021	**(c) unidirectional light**

Exam trap: (b) The options of this paper are not preserved; the answer it wants is **coaxial**. (d) The printed key says "none of these", but (a) and (b) are advantages; one-way light (two fibres for two-way traffic) is the textbook **disadvantage**, so answer (c).

19 Optical fibre

Q56: Explain the structure and characteristics of fibre-optic cable, its advantages and disadvantages, and how data travels through it.
✎ **Asked in:** 17th NTRCA ICT (2023) · 13th NTRCA CSE (2016)



Structure: glass **core** (9 µm single-mode, 50/62.5 µm multimode), **cladding** with a lower refractive index, buffer coating, strength members (Kevlar), outer jacket. **How data travels:** a laser / LED sends light pulses; light hitting the core–cladding boundary above the **critical angle** is totally reflected back, so it stays in the core.

Advantages: huge bandwidth, very low loss, immune to EMI, light and thin, no sparks, hard to tap. **Disadvantages:** costly installation and splicing, fragile, light travels one way, needs O/E conversion.

Q57: What are the main requirements of optical fibre splicing?
✎ **Asked in:** BPDB AE(CSE) 2021

① exact **core alignment** (lateral, angular, gap) ② clean, flat, perpendicular **cleave** ③ **low insertion loss** (fusion ≈ 0.05–0.1 dB) and low back-reflection ④ mechanical strength: protective sleeve / splice tray ⑤ dust-free, well-matched fibres. Fusion splicing (arc-welded) beats mechanical splicing on loss.

Q58: Single-mode fibre 0.35 dB/km, splitter loss 14 dB, two splices of 0.1 dB, two connectors of 0.75 dB, transmitter 5 dBm, receiver sensitivity −14 dBm. Find the maximum OLT–ONU fibre length.
✎ **Asked in:** IBBL SO(Network) 2025

→ budget = 5 − (−14) = 19 dB · fixed losses = 14 + 0.2 + 1.5 = 15.7 dB · left for fibre = 3.3 dB → L = 3.3 / 0.35 = **9.43 km**

> **Q59:** A 1550-nm link is 50 km long without repeaters; the single-mode fibre loses 0.2 dB/km and is joined every 2 km with connectors of 0.5 dB each. Find the minimum average launch power for 10 µW at the receiver.
> ✎ **Asked in:** BTCL AM(Technical) 2021

→ fibre 50 × 0.2 = 10 dB · joints every 2 km = 24 × 0.5 = 12 dB · total 22 dB · 10 µW = −20 dBm → launch = −20 + 22 = **+2 dBm ≈ 1.58 mW**

> **Exam trap:** 25 segments of 2 km have **24** joints between them (the end connectors belong to the transmitter and receiver). Work in dB and dBm, then convert once at the end.

20 Satellite, submarine cable, mmWave

> **Q60:** What are optical cable and satellite transmission? Differentiate them. Name Bangladesh's submarine cables. Why is fibre used under the sea instead of satellite, and why do submarine cables get damaged?
> ✎ **Asked in:** RAKUB ANSE 2023 · BSCPLC AM(Engineering) 2024

Fibre: light in glass, terabits, ≈ 5 µs/km delay, cheap per bit, repairs need a ship. **Satellite**: radio via a GEO satellite at 35,786 km, **≈ 240–270 ms one-way delay**, limited bandwidth, weather fade, but covers remote areas, ships and planes. Hence submarine fibre carries the bulk of international traffic. Bangladesh: **SEA-ME-WE 4** (Cox's Bazar), **SEA-ME-WE 5** (Kuakata), **SEA-ME-WE 6** (Cox's Bazar). Damage: ship anchors and fishing trawls (most faults), earthquakes and undersea landslides, abrasion on rocks, shark bites in old cables.

> **Q61:** What is unguided media? Write the features of satellite communication.
> ✎ **Asked in:** 18th NTRCA School-2 (2024)

Unguided media carry electromagnetic waves through air / space without a conductor (radio, microwave, infrared, satellite). **Satellite features:** wide coverage (3 GEO satellites cover the globe), distance-insensitive cost, broadcast to many receivers, uplink/downlink on different frequencies (C, Ku, Ka band), long delay, rain fade, very high launch cost.

MCQ	Answer
a. VSAT stands for: (a) Virtual Small Aperture Satellite (b) Very Small Aperture Terminal (c) Very Small Application Terminal (d) Vertical Satellite ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(b) Very Small Aperture Terminal**
b. Online banking from a plane or a ship needs: (a) fibre (b) satellite (c) microwave (d) coaxial ✎ **Asked in:** BB AD(General) 2022	**(b) satellite**
c. Bluetooth uses: (a) infrared (b) radio waves (2.4 GHz) (c) laser (d) microwave dish ✎ **Asked in:** BB AD(General) 2022	**(b) radio waves (2.4 GHz)**

Q62: What is the purpose and primary application of millimetre waves (mmWave) in IoT ecosystems?
✎ **Asked in:** Petrobangla AM(CSE) 2024

mmWave = 30–300 GHz (5G uses 24–100 GHz). **Purpose:** very wide channels → multi-Gbps rates and about 1 ms latency for dense device clusters; small antennas allow beam-forming arrays. **Applications:** 5G fixed wireless and backhaul, industrial IoT / smart factories, AR/VR, connected vehicles (V2X), high-resolution radar sensing. **Limit:** short range, blocked by walls and rain, so it needs dense small cells.

Satellite orbits at a glance

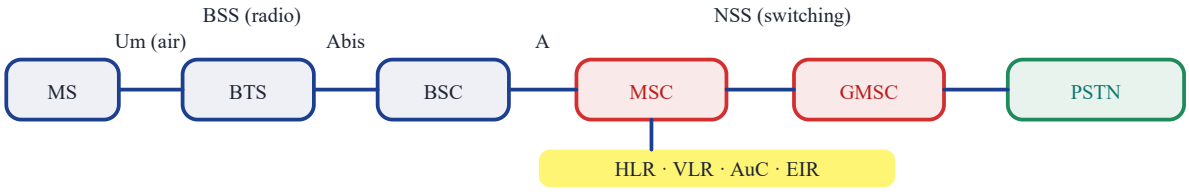
Orbit	Height	One-way delay	Examples
LEO	≈ 500–2000 km	≈ 5–20 ms	Starlink, Iridium, imaging satellites
MEO	≈ 2000–35,786 km	≈ 50–150 ms	GPS (≈ 20,200 km), O3b
GEO	**35,786 km**	**≈ 240–270 ms**	TV broadcast, VSAT, Bangabandhu Satellite-1 (119.1° E)

Memory trick: **GEO** = **G**oes round **E**xactly **O**nce a day: it stays fixed over one point, so dishes never move, but every hop costs about a quarter of a second.

Viva rapid-fire: Satellite vs submarine cable → delay, capacity, cost (Q60) · SEA-ME-WE = South-East Asia–Middle East–Western Europe · landing stations: **SMW-4 Cox's Bazar, SMW-5 Kuakata (Patuakhali)** · fibre vs Ethernet copper: fibre for long / noisy runs, copper for short desk links; both lose data only through errors, not by design · light stays in fibre by total internal reflection (core n > cladding n) · first fibre telephone links: 1977, USA. ✎ **Source:** bank IT viva (candidates' accounts) · Rupali Bank ANE viva

21 Mobile systems: GSM, GPRS, LTE

Q63: Draw the GSM architecture. Write the full meaning of GSM and CDMA. What are the functions of Node B and GGSN?
✎ **Asked in:** Teletalk Executive(Technical) 2015 · Teletalk Executive 2019 · Combined Bank SO(IT) 2020



GSM = Global System for Mobile communications; **CDMA** = Code Division Multiple Access. **Node B**: the 3G (UMTS) base station: WCDMA radio, modulation, power control, controlled by the RNC. **GGSN** (Gateway GPRS Support Node): gateway between the mobile packet core and external networks (Internet); assigns IP addresses, routes packets to the right SGSN, applies policy, firewall and charging.

Q64: What is LTE (full meaning)? Name its advanced technology. Explain LTE and GSM and the differences between them; describe LTE radio technology; list the LTE network elements.

✎ **Asked in:** BTRC AD 2021 · BTRC DAD 2021 · BTRC AD(Technical) 2019 · Combined Bank Officer(IT) 2020

LTE = Long Term Evolution (4G, 3GPP Release 8); advanced forms **LTE-Advanced** (Rel. 10: carrier aggregation, 8×8 MIMO) and LTE-Advanced Pro. Radio: **OFDMA downlink, SC-FDMA uplink**, MIMO, 1.4–20 MHz channels, all-IP. **Elements:** UE, **eNodeB** (E-UTRAN), and the EPC: **MME** (signalling, mobility), **S-GW**, **P-GW** (Internet gateway, IP address), **HSS** (subscriber database), PCRF (policy).

Point	GSM (2G)	LTE (4G)
Access	TDMA + FDMA	OFDMA / SC-FDMA
Switching	circuit (voice)	packet, all-IP (VoLTE)
Channel	200 kHz	up to 20 MHz
Data rate	9.6 kbps (GPRS ~ 114)	100+ Mbps
Base station	BTS + BSC	eNodeB only

Q65: What is GPRS? Write its features and advantages.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

GPRS (General Packet Radio Service, "2.5G") adds packet switching to GSM: SGSN + GGSN nodes, up to 171 kbps theoretically (≈ 40–114 kbps real). Features: always-on, billed by data volume, timeslots shared by many users, IP-based (WAP, email, MMS). Advantages: efficient use of radio, instant access, the base for EDGE and 3G cores.

Q66: What is the function of TCSM? What equipment is needed for VoIP? What is a call drop and what causes it? What are GPS, VoIP, CDMA and GSM, and why are they used?

✎ **Asked in:** BTRC DAD 2021 · BTRC DAD(Technical) 2019

1. **TCSM** (transcoder and sub-multiplexer): converts the 13-kbps GSM speech (16 kbps with overhead) to 64-kbps PCM for the MSC, and packs four 16-kbps channels into one 64-kbps timeslot, saving transmission lines.
2. **VoIP equipment:** IP phone or softphone / ATA adapter, broadband IP network with QoS, SIP server / IP-PBX (softswitch), VoIP gateway to the PSTN, codecs (G.711, G.729), router / firewall.
3. **Call drop** = a call cut off before either user ends it. Causes: weak coverage, handover failure, interference, network congestion, faulty BTS / backhaul, high speed of the user, handset fault.
4. **GPS**: satellite positioning (navigation, timing) · **VoIP**: voice over IP networks, cheap calls · **CDMA**: users share the band by codes · **GSM**: 2G digital cellular standard for voice and SMS.

Q67: Discuss the advantages and disadvantages of TDMA and CDMA.

✎ **Asked in:** 46th BCS ICT

	TDMA	CDMA
Advantages	simple, cheap handsets, battery saving (radio off in other slots), easy hand-off	more users per MHz, soft hand-off, resists interference and tapping, frequency reuse 1
Disadvantages	needs strict sync and guard time, fixed slots wasted, multipath ISI	needs tight power control (near–far problem), complex receivers, capacity falls as load rises

Q68: Identify the differences between 1G, 2G, 3G and 5G networks. Write the disadvantages of 5G in terms of initial cost and battery drainage.

✎ **Asked in:** Fisheries & Livestock Instructor(CS) 2023 · Rupali Bank ANE 2021

Gen	Technology	Signal / access	Speed	Service
1G	AMPS	analog, FDMA	voice only	voice
2G	GSM, IS-95	digital, TDMA / CDMA	9.6–64 kbps	voice, SMS
3G	UMTS / WCDMA	WCDMA	≈ 2 Mbps+	video call, mobile Internet
5G	NR	OFDM, mmWave, massive MIMO	up to 20 Gbps, ≈ 1 ms	eMBB, IoT, URLLC

5G disadvantages: initial cost: dense small cells, fibre backhaul and new spectrum make roll-out very expensive; **battery drainage**: wide-band radios and many antennas use more power and heat, so phones drain faster, especially while searching between 4G and 5G.

Q69: Write the differences between a digital telephone and a mobile phone. What is a modem? Write its uses.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

Digital (fixed) telephone: wired to the exchange, fixed location, digital switching (ISDN / PCM), very stable. **Mobile phone**: wireless cellular link to a base station, moves with hand-over, needs a SIM and battery, adds SMS and data. **Modem** (modulator–demodulator) converts digital data to an analog signal for the line and back. Uses: dial-up / DSL / cable Internet, fax, GSM modems for SMS alerts, POS terminals, remote monitoring.

Q70: Explain the parity method for error detection. Write the bit strings of "Delta" using ASCII.

✎ **Asked in:** BB AME 2023

The sender adds **one parity bit** so the count of 1s is even (even parity) or odd (odd parity); the receiver recounts. It detects any **odd number** of flipped bits, never an even number, and cannot correct.

Char	ASCII	7-bit	even parity (MSB)
D	68	1000100	01000100
e	101	1100101	01100101
l	108	1101100	01101100
t	116	1110100	01110100
a	97	1100001	**11100001**

MCQ	Answer
a. Two-tier client/server: application and UI run on the: (a) server side (b) host side (c) modulation side (d) none (client side) ✎ **Asked in:** Combined Bank ADA 2021	**(d) none (client side)**
b. Asymmetric encryption: the key that encrypts is the: (a) private key (b) encryption key (c) public key (d) modulation key ✎ **Asked in:** Combined Bank ADA 2021	**(c) public key**
c. Switching delayed because data is stored in and read from RAM: (a) space-division (b) time-division (c) virtual (d) packet ✎ **Asked in:** Combined Bank SO(IT) 2021	**(b) time-division**

Viva rapid-fire: Name some cables → twisted pair (UTP/STP), coaxial, optical fibre · what is a medium → the physical path between sender and receiver; guided or unguided · SIM → Subscriber Identity Module: a smart card holding the IMSI and the key Ki that the network uses to authenticate the subscriber · a digital signal → discrete levels (draw a square wave) · polarisation → the direction of the electric field of a wave (vertical, horizontal, circular); antennas must match it · Ethernet sends digital data as voltage levels (line-coded), not an analog carrier. ✎ **Source:** Combined Bank Officer (IT) 2026 viva · BPSC viva 2023 · bank IT viva · Rupali Bank ANE viva

☾ Night before the exam

Formulas: Nyquist rate = $2f_{max}$ ($\omega = 2\pi f$!)

Nyquist $R = 2B \log_2 L$ · Shannon $C = B \log_2(1+SNR)$

$SNR = 10^{dB/10}$ · 3 dB = ×2 · 10 dB = ×10

bps = baud × $\log_2 L$ · QAM-16 = 4 bits/baud

FDM $B = NB + (N-1)G$ · TDM out = n × in

AM $B = 2f_m$ · FM $B = 2(\Delta f + f_m)$, $\beta = \Delta f / f_m$

PCM $R = n \cdot f_s$ · SQNR = 1.76 + 6.02n dB

$H = -\Sigma p \log_2 p$ · $R = r \cdot H$ · $\lambda = c/f$

One-liners: TDM = time slots, digital, needs sync · FDM = bands, analog, guard bands · WDM = light, fibre

GSM → GMSK · most efficient → 16-QAM · DM noise → slope overload + granular

VHF 30–300 MHz · DTMF 697 + 1477 = "3"

Manchester = mid-bit transition · 4B/5B ≤ 3 zeros

fibre windows 850 / 1310 / **1550** nm · SMW-4, 5, 6

LTE: OFDMA ↓ SC-FDMA ↑ · GGSN = Internet gateway

Which formula for which wording

The question says …	Use	See
"guard band", "minimum bandwidth of the link"	NB + (N − 1)G	Q4
"unit is 1 bit", "frame duration / slot"	frame = input unit time; slot = frame ÷ n	Q8–Q11
"SNR", "noisy channel", "capacity"	Shannon, SNR as a ratio	Q21–Q23
"noiseless", "signal levels"	Nyquist $2B \log_2 L$	Q24
"sin(ωt)", "Nyquist rate / interval"	$f = \omega/2\pi$, then 2f	Q19
"step size", "SQNR"	$(A^2/2) \div (\Delta^2/12)$	Q39
"probabilities", "information rate"	H, then R = r·H	Q44, Q45
"dB/km", "sensitivity", "launch power"	power budget in dB	Q58, Q59

All the tricks: Turns-in-Time / Floors / Wavelengths · fence posts (N − 1 guard bands) · Shannon Says noise · AMRN (why modulate) · small step can't climb · halves give whole bits · bands start with 3.

Top traps: ω vs f in sin(2100πt) · SNR ratio vs dB · "150% higher" = 2.5× · AM bandwidth ignores the carrier · a circuit gets one TDM slot · latch vs flip-flop · 24 joints in 50 km at 2-km spacing.

✓ Practice MCQs (asked ones first)

1. Main disadvantage of TDM over FDM: ✎ 49th BCS ICT (2025)

(a) higher bandwidth (b) noise immunity
(c) complex sync (d) limited frequency

2. Which multiplexing transmits digital signals? ✎ BPSC AME 2019

(a) WDM (b) FDM
(c) TDM (d) all of these

3. SNR = 63, B = 1 MHz: bit rate = ✎ BPSC AME 2019

(a) 4 Mbps (b) 6 Mbps
(c) 8 Mbps (d) 10 Mbps

4. Combines signals by assigning time slots: ✎ 49th BCS CSE (2025)

(a) TDM (b) FDM
(c) WDM (d) SDM

5. AM: carrier 300 kHz, message 3 kHz, bandwidth: ✎ DoICT ANE 2019

(a) 300 Hz (b) 3 kHz
(c) 600 Hz (d) 6 kHz

6. VSAT: ✎ BDCCL AM(Transmission) 2022

(a) Very Small Aperture Terminal (b) Virtual Small Aperture Satellite
(c) Very Small Application Terminal (d) Vertical Satellite

7. Codec digitising analog signals uses:

(a) demodulation (b) PCM
(c) CDMA (d) NRZ

8. LTE means:

(a) Long Term Evaluation (b) Long Time Evaluation
(c) Long Terminal Evolution (d) Long Term Evolution

9. Nyquist rate of sin(2100πt):

(a) 1050 Hz (b) 4200 Hz
(c) 2100 Hz (d) 6600 Hz

10. Bits per baud in 64-QAM:

(a) 6 (b) 4
(c) 8 (d) 64

Answer key → 1-c · 2-c · 3-b · 4-a · 5-d · 6-a · 7-b · 8-d · 9-c · 10-a

Written question: What is multiplexing? Write the differences between TDM and FDM, and find the minimum link bandwidth for five 100-kHz channels with 10-kHz guard bands. 10 marks

✎ **Asked in:** 45th BCS CSE (2024) · BPSC AP 2021 · Combined Bank SO(IT) 2026

1. Definition: several signals share one link; MUX at the sender, DEMUX at the receiver; why: link capacity ≫ one signal 2
2. Diagram: MUX → one link → DEMUX, TDM slots vs FDM bands 2
3. Table (at least 5 points): shares time vs frequency, digital vs analog, sync bits vs guard bands, needs synchronisation vs filters, examples E1 vs radio 4
4. B = 5 × 100 + (5 − 1) × 10 = **540 kHz**, with the four guard bands drawn 2

Before the exam ✓

- ✓ TDM vs FDM vs WDM table
- ✓ TDM frame: duration, slot, rate
- ✓ Shannon with SNR in dB
- ✓ Nyquist rate from ω
- ✓ PCM blocks + pros/cons
- ✓ AM / FM bandwidth, β